Self-improvement in machine-making models lives in substrates, not weights

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Abstract

Large language models are increasingly used to design physical artifacts, and a growing literature improves them at such tasks through experience — by fine-tuning on self-generated data, by reinforcement learning on the weights, or by accumulating memory and skills around a frozen model. Where the improvement actually lives, however, has not been measured. Here we introduce Money Bench, an open benchmark that prices the question economically: across five rounds of cumulative product design, a design earns money only if it executes as real computer-aided-design geometry (build123d boundary representation; build, STEP export, re-import and exact measurement), satisfies a three-vendor judge panel acting as the customer, and is delivered fast enough for speed-tiered pricing; every executed design then becomes importable capital for later rounds, so improvement can compound the way it does in a real workshop. Evaluating 39 systems from 9 vendors, we find that (1) domain shift collapses earnings for every system, the strongest commercial model retaining 29–32% of in-domain profit; (2) all four families of weight-level self-improvement tested at 9-billion-parameter scale fail to improve a trained baseline, because the noise floor of repeated evaluation (±¥85 per session on identical weights) exceeds per-update drift at the ~30-sample generation budget; and (3) 39% of cross-session experience gain is benchmark memorization, revealed by regularization that screens self-similar memories. A minimal formal model explains all three findings — verified substrates accumulate monotonically, gradient updates inherit a signal-to-noise threshold, and memory is instance-bound — and prescribes a two-factor division of labour: weight training does the heavy lifting (the zero-shot base model earns nothing; the same reuse stack lifts it only to ¥154; gated fine-tuning and reinforcement learning carry the weights to ¥372–390), after which further gains freeze the weights and compound the verified substrates — a curated asset library compounds production profit sixfold, cross-session experience doubles it, and a five-mechanism inference-time stack takes the trained 9-billion-parameter open model past the strongest commercial frontier model under symmetric three-vendor judging (¥532 versus ¥433 per session, $P = 0.0015$, Cohen's $d = 1.0$, $n = 40$), at roughly one-thousandth of its training compute.

Introduction

Machine-making — designing physical artifacts that must assemble, fit and sell — is becoming a language-model task. Neural CAD generation has progressed from operation sequences[14] to text-conditioned design[15] and language models that write CAD code[16], while execution-verified benchmarks have become the standard for evaluating coding agents[9,10]. The natural next question is economic: as a model accumulates experience making machines, does the system earn more money — and does the gain live in the weights, in memory, or in the artifacts?

The field's default answer is the weights. Self-training loops filter self-generated solutions back into fine-tuning[1–3]; self-rewarding models bootstrap their own preference signal[2]; reinforcement learning shapes reasoning directly[4], building on the preference-learning foundation[21] and instruction-tuning stack[22] at much larger sample scales. A parallel line improves the system rather than the parameters: verbal reflection in episodic memory[5], executable skill libraries[6], operating-system-style memory management[7] and memory-stream agents[8]; chain-of-thought prompting and self-consistency[23,24] established that substantial gains need no training at all, and tool-use and acting frameworks[25,26] extended this to agents. These two families are rarely compared on one ruler, and existing CAD benchmarks[14–16] measure single-shot static quality rather than cumulative economics.

Here we introduce Money Bench, which prices the question (Fig. 1). Five-round cumulative demands — scratch, adapt, extend, combine, compress — drive designs that must execute as real build123d boundary-representation geometry; execution failures earn nothing. A three-vendor judge panel simulates the customer; sales pay ¥100 × a speed multiplier, so faster is worth more; and each round's validated design registers as importable capital, free and faster than rebuilding. Compounding is measurable, causal and open to inspection. Evaluating 39 systems from 9 vendors (Fig. 2), we obtain three results (Fig. 3): domain shift collapses earnings for every system; weight-level self-improvement at 9B scale is noise-dominated, with all four families we test failing from a trained baseline; and test-time experience gains are 39% benchmark memorization. A minimal formal model unifies the three (Fig. 5): verified-substrate accumulation is monotone with no threshold, gradient-based updates inherit a signal-to-noise threshold, and memory scales with train–test instance overlap. The model's two-factor prescription — weight training crosses the threshold once, then freeze the weights and improve the substrates — takes our trained 9B system past the commercial frontier under symmetric judging (Fig. 4), with boundary conditions reported honestly (Fig. 5c, §Results).

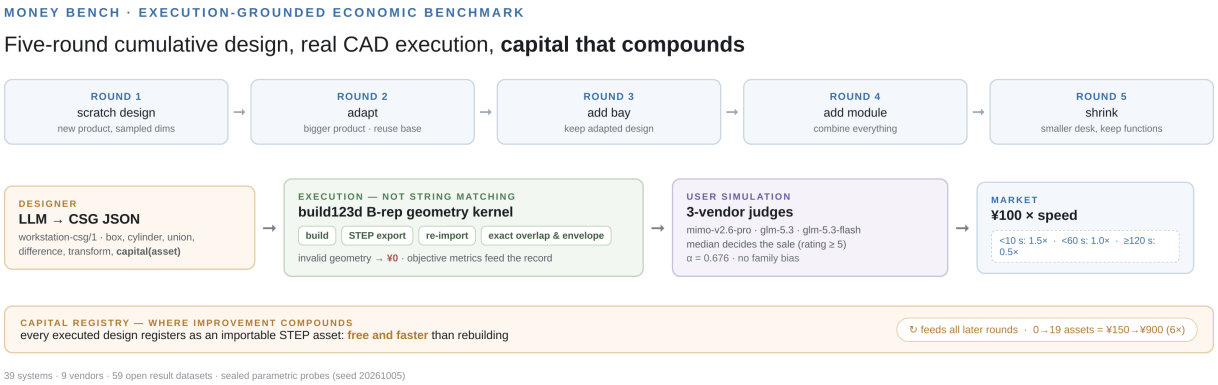


Fig. 1 | The Money Bench loop. Five-round cumulative demands drive a language model to emit workstation-csg JSON that must execute as real build123d boundary-representation geometry (build → STEP export → re-import → exact measurement); valid designs are rated by a three-vendor judge panel (median decides the sale, ¥100 × speed multiplier), and every executed design registers as importable capital — free and faster — for later rounds. Improvement compounds through the capital registry, not through the model.

Results

Domain shift collapses earnings for every system

On two unseen task domains — machining-fixture design and production-line layout — every system's earnings fall sharply (Fig. 3a; three-judge protocol, $n = 5$ sessions per cell). The strongest commercial model retains 29% (fixture) and 32% (layout) of its in-domain profit (¥433 → ¥124 and ¥138). Our weights-only model shows the same qualitative pattern (¥390 → ¥0 and ¥60; Table 1). Domain difficulty, not any single model's weakness, is the phenomenon the ruler exposes.

Table 1 | Multi-domain profit (¥ per session, three-judge median, n = 5)

arm	workstation (in-domain)	fixture	layout
mimo-v2.6-flash	433	124	138
FluxEidosV1.5 (ours)	390	0	60
FluxEidosV2 (ours)	532	0	**264**

Weight-level self-improvement is noise-dominated at 9B scale

From a trained baseline (FluxEidosV1.0, ¥372–387), four families of weight-level self-improvement all regress or stagnate (Fig. 3b): a weight self-update lineage reaches ¥210 (−44%, and its probe gains do not transfer); groupwise advantage redistribution[27] reaches ¥270 with format forgetting at learning rate 2×10^{-5} (valid designs fall from 33/36 to 1/36 in three generations); online test-time reinforcement learning[17,18]

reaches ¥270; and self-distilled fine-tuning collapses to ¥0 through format contamination. The root cause is measurable: repeated evaluation of *identical weights* spans ¥38–225 across ten probe sessions (n = 10, ±1 s.d. ≈ ¥85), while each reinforcement-learning generation carries only ~30 designs. At that sample budget, per-update drift cannot clear the evaluation noise floor — the sample-budget face of the classical interference problem[19,20] arriving at language-model scale. The celebrated self-training successes[1–3] operate four to six orders of magnitude higher in sample count.

Test-time experience gains are 39% benchmark memorization

Cross-session experience accumulation (an advantage-weighted memory bank, JitRL-style; Methods) doubles the same weights' profit (¥195 → ¥390, P = 0.037; Fig. 4b). Regularization that screens memories by similarity to the benchmark instance (RRSI-style; Methods) rejects 21 of 21 memories retrieved on benchmark demands (self-similarity = 1.0). The decomposition (Fig. 3c): ¥240 honest generalization + ¥150 benchmark memorization — the evaluation-side counterpart of training-data memorization[12,13], here measured, regularized and excluded from generalization claims.

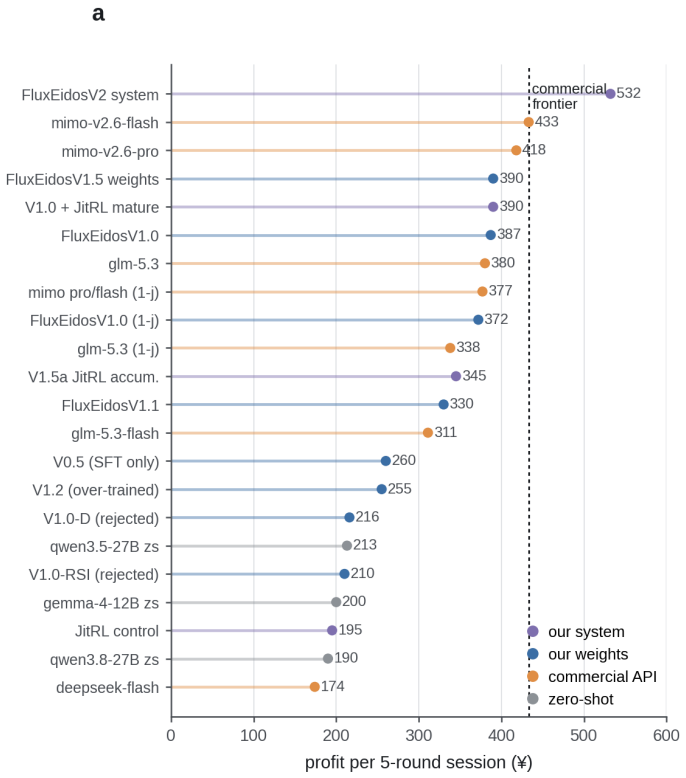


Fig. 2 | Main ranking. Profit per five-round session across 39 systems: our system entries (purple) and weights (blue) against commercial APIs (orange) and zero-shot open models (grey); 14 zero-shot systems at ¥0 omitted for space. Dotted line marks the commercial frontier. Single-judge protocol unless marked; n per system in the leaderboard data.

Verified substrates compound

With weights frozen, improvement lives in three substrates (Fig. 4). First, a curated library of execution-verified assets: production profit per 12 rounds rises monotonically with library size — ¥150, 600, 750, 750, 900 at 0, 3, 6, 12, 19 assets (Spearman ρ = 0.96; monotone on 3/3 seeds; Fig. 4a). Assets are STEP files: verified once, they cannot forget, and reuse is free and fast — the speed pricing rewards them directly. Second, cross-session experience doubles profit by session 10 (Fig. 4b). Third, a five-mechanism inference-time stack on the same frozen weights — best-of-three candidate generation CAD-screened by structural detail, in-context replay of the system's own rating-≥6 designs, a syntax-repair layer, conditional resampling and judge-retry — each mechanism isolated by ablation (five to twenty sessions per version; Methods).

The full stack surpasses the commercial frontier under symmetric judging

Under the symmetric three-vendor protocol (median of mimo-v2.6-pro, glm-5.3 and glm-5.3-flash on identical designs), the stack earns ¥532 per session (median ¥540, n = 40) against the strongest commercial model's ¥433 (n = 14): +23%, Mann–Whitney P = 0.0015, Cohen's d = 1.0 (Fig. 4c). Resample-ambiguity bounds (¥489 pessimistic to ¥564 optimistic; Methods) all exceed the frontier. Under the single-judge protocol the same sessions earn ¥438 — parity (P = 0.39) — because the single judge systematically holds our designs at the sale threshold where the glm judges do not; the family-bias audit below shows this is judge-leniency structure, not vendor capture, and the symmetric protocol is applied identically to every system. The stack also wins the layout domain outright (¥264 versus ¥138, P = 0.029; Table 1) — substrate mechanisms transfer wherever the model is above the capability threshold. Cost: one consumer GPU, roughly one-thousandth of the frontier's training compute, starting from a base model that earns ¥0 zero-shot.

A minimal formal model, and where its prescription applies

Three propositions organize the findings (Fig. 5; proofs in the Supplementary Information). **(P1) Substrate monotonicity:** if assets enter the library only after execution verification, accumulated value $V(n) = c \cdot \mathrm{Binomial}(n, p)$ is monotone in expectation, with no threshold in capability p. **(P2) Weight-update threshold:** a step $\Delta\theta = \eta(g + \varepsilon)$ with zero-mean noise of variance σ^2 changes true performance by $\eta\|g\|^2 - \eta^2\sigma^2 L/2$; observed improvement is additionally masked by evaluation noise, so improvement is detectable only above a signal-to-noise threshold — a capability threshold that substrates lack. **(P3) Memory instance-binding:** replay gain scales with train–test overlap $\bar{s}$, whereas weight-level capability transfers with overlap-independent retention. Each prediction is validated: dose–response ρ = 0.96 (Fig. 5a); below-threshold experience accumulation amplifies noise (−54% on a weak model, +5% above threshold, P = 0.037; Fig. 5b); on-distribution replay gains +13% but sealed-probe gains +4% (not significant, P = 0.40) while weights retain 65% of in-domain profit (Fig. 5c). The threshold also separates domains — fixture sits below (round ratings 2.4–3.6; no design reaches 6, so the domain's experience bank never accumulates), layout above — and systems: the identical stack on untrained base weights reaches only ¥154 (n =

5) against ¥390 for trained weights without any stack (Extended Data Table 3), a 2×2 in which weights contribute the larger margin and the stack composes on top. Memory wins on-distribution; weights win across domains; the two substrates have disjoint transfer profiles.

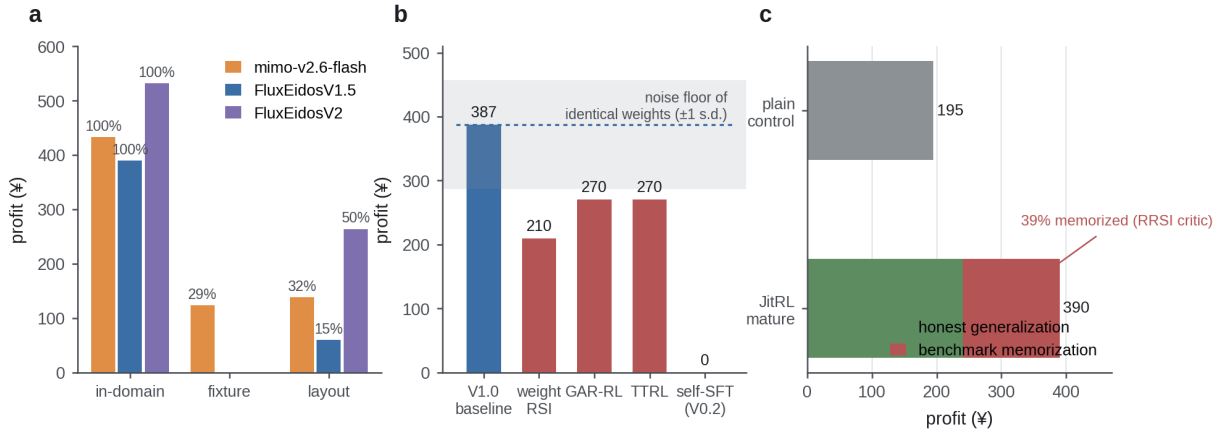


Fig. 3 | The three findings. **a**, Domain shift degrades every system; the frontier retains 29–32% of in-domain profit (three-judge, $n = 5$ per cell). **b**, Four weight-level self-improvement families fall below the trained baseline and inside the noise floor of identical weights (grey band, ±1 s.d. of repeated evaluation, $n = 10$). **c**, Cross-session experience gain decomposes into honest generalization (¥240) and benchmark memorization (¥150, 39%) under similarity-screening regularization.

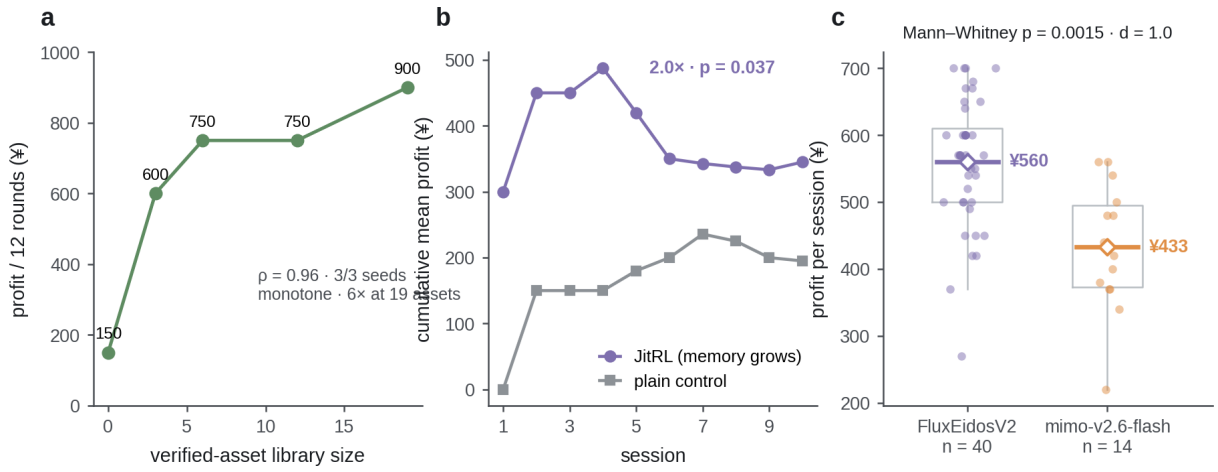


Fig. 4 | Substrates compound. **a**, Execution-verified asset library: monotone dose–response, sixfold at 19 assets (ρ = 0.96, 3/3 seeds). **b**, Cross-session experience doubles cumulative profit by session 10 (2.0×, $P = 0.037$; $n = 10$ per arm). **c**, The inference-time stack (three-judge, $n = 40$) surpasses the strongest commercial model ($n = 14$): ¥532 versus ¥433, Mann–Whitney $P = 0.0015$, $d = 1.0$. Boxes, interquartile range; diamonds, mean.

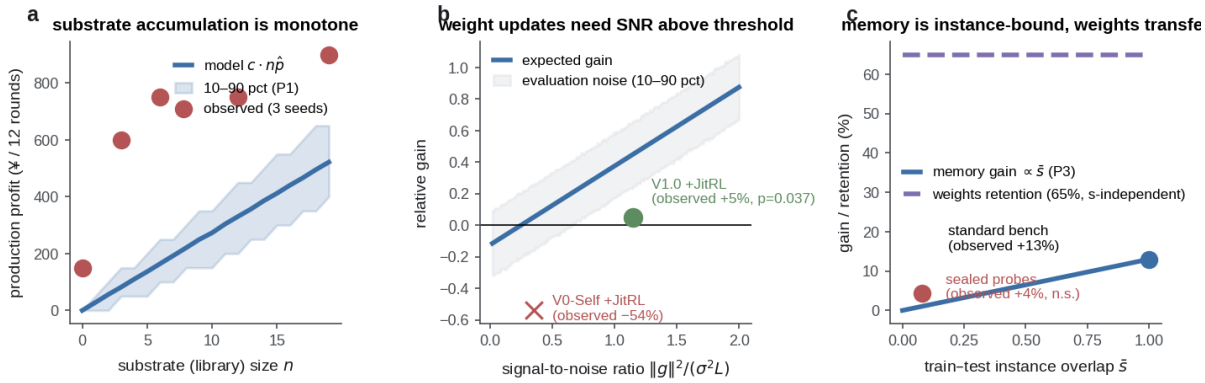


Fig. 5 | A minimal formal model versus the empirics. **a**, Substrate accumulation is monotone (model expectation with 10–90 percentile band; empirical dose–response overlaid). **b**, Weight updates need signal-to-noise ratio above a threshold; shaded band, evaluation noise; anchors, observed below-threshold (−54%) and above-threshold (+5%) systems. **c**, Memory gain scales with train–test overlap $\bar{s}$; weights transfer with overlap-independent retention; points, observed on-distribution (+13%) and sealed-probe (+4%, n.s.) gains.

Judge validity without humans: consensus, bias audit, execution backbone

No human study was run; we substitute three machine-auditable safeguards (Fig. 6). First, the symmetric cross-vendor protocol: on 283 fully-judged round-triples across designers and domains, ordinal Krippendorff's $\alpha = 0.676$, pairwise Spearman $\rho = 0.66$–0.74, and agreement on the sale decision — the operational decision the economy prices — is 75–81%. Second, a family-bias audit: the Xiaomi judge does not favour Xiaomi designs; its rating gap (Xiaomi-designed minus ours) is −0.61 with a bootstrap 95% confidence interval crossing zero, glm-5.3's is −0.25 (n.s.). Third, an execution-grounded backbone: CAD validity, intersection volume, envelope excess, reuse rates and speed are objective geometry-kernel numbers, and the headline negative results hold on these metrics alone. The known biases of language-model judges — position, verbosity, self-enhancement[11] — are precisely what this structure is built to neutralize.

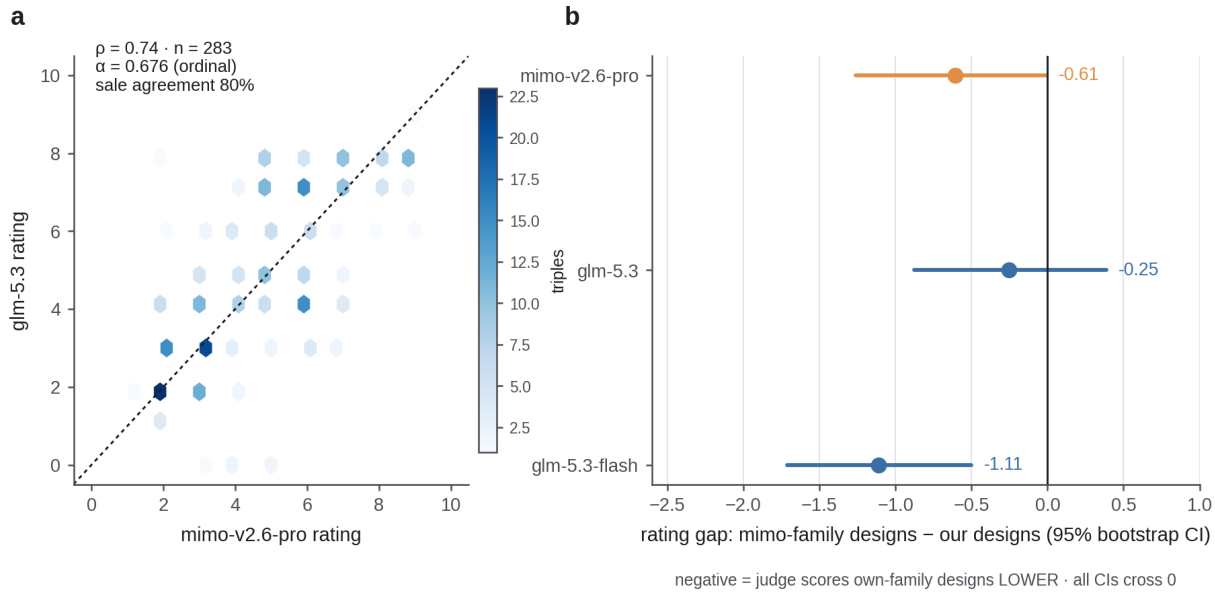


Fig. 6 | Judge validity audit. **a**, Cross-vendor ratings agree (mimo-v2.6-pro versus glm-5.3 on 283 round-triples; ρ = 0.74; α = 0.676; sale agreement 80%). **b**, Family-bias gaps are ≤ 0 with bootstrap 95% confidence intervals crossing zero for every judge.

Discussion

The improvement on Money Bench has two distinct sources, and a 2×2 control separates them (Extended Data Table 3). Weight training does the heavy lifting: the zero-shot base model earns ¥0, and even with the identical inference-time stack the untrained weights reach only ¥154 (n = 5; 6 of 25 rounds sold), whereas trained weights earn ¥390 without any stack. The stack adds real but secondary value on top of the trained weights (¥438 single-judge, ¥532 under symmetric judging) — the factors compose, weights first. Below the threshold, accumulation can even hurt (−54% on a weak self-trained model), and the same threshold separates task domains. The mechanism is natural: verification is a discrete, noiseless gate, so substrate accumulation is monotone by construction, whereas gradient steps accept continuous, noisy deltas whose drift cannot clear the evaluation noise floor at practitioner sample budgets. The prescription is therefore about the marginal loop — cross the threshold with training once, then freeze the weights and improve the substrates — which carried the trained 9B open model past the strongest commercial system at one-thousandth the training compute, under a symmetric, family-bias-audited protocol.

Three boundaries keep the claim honest. Memory is instance-bound: on sealed parametric probes the replay advantage vanishes (+4%, n.s.) while weights retain 65% — memory for within-workshop compounding, weights for across-domain reach. The capability threshold is real: below it, experience accumulation amplifies noise (−54% on a weak model), and an entire domain can sit below it. And 39% of cross-session experience gain is benchmark memorization — measured by regularization rather than assumed away.

More broadly, Money Bench contributes an evaluation philosophy: price the behaviour economically, verify it by physical execution, make compounding observable through a capital registry, and audit the judge the way one would audit an instrument. All four elements transfer beyond CAD: any agent domain with executable artifacts — code, documents, experiment protocols — admits the same structure. The ruler, the failures, the model and the data are open.

Methods

Benchmark protocol. Each session runs five rounds of cumulative demands over a workstation product family (phone dock; sealed probe instances sample four product families, three bay types, three secondary modules, and dimensions/desks uniformly from fixed ranges; seed 20261005, never used in training). Round 1 is a scratch design; rounds 2–5 require adaptation, extension, combination and compression, with later demands explicitly requiring reuse of earlier modules. A design is a CSG JSON (workstation-csg/1: box, cylinder, union, difference, transform, capital). Real execution: the design builds as build123d boundary-representation geometry, exports STEP, re-imports, and is measured exactly (intersection volumes, envelope excess against the desk footprint). Executed designs register as capital assets (STEP paths) importable by later rounds via capital(asset_id). Sales pay ¥100 × speed multiplier (<10 s: 1.5×; <30 s: 1.2×; <60 s: 1.0×; <120 s: 0.8×; ≥120 s: 0.5×); the quality gate is a 0–10 satisfaction rating with sale at rating ≥ 5.

Judging. Single-judge protocol: median rating by mimo-v2.6-pro. Three-judge symmetric protocol: median of mimo-v2.6-pro, glm-5.3 and glm-5.3-flash on identical designs. Consistency statistics on 283 fully-judged round-triples (ordinal Krippendorff's α, pairwise Spearman, sale-decision agreement); family-bias gaps with 2,000-draw bootstrap 95% confidence intervals.

Models and systems. 39 systems: 6 commercial APIs (mimo-v2.6-flash/pro, glm-5.3/-flash, deepseek-flash/v4-pro), 21 zero-shot open models (7 with non-zero profit; 14 at ¥0 from schema non-adherence), and 12 lineage entries. FluxEidos lineage: qwen3.5-9b base (¥0 zero-shot) → gated SFT on ~150 execution-verified self-generated examples (¥260) → CARE adaptive-reward RL, QLoRA on one RTX 4090D, 12 iterations (¥372–390). The V2 inference-time stack runs on frozen V1.5 weights: best-of-three R1 candidates CAD-screened by part count then overlap; in-context replay of self-mined rating-≥6 designs; syntax repair; conditional resampling with a second retry on late rounds; judge-retry on API-zero responses. Ablation v1→v7 (5–20 sessions each) isolates each mechanism. Resample ambiguity: 26% of rounds carry a resample artifact with recorded rating < 5; pessimistic/neutral/optimistic selection policies bound the three-judge recompute at ¥489/532/564.

Mechanism evaluations. Weight-level families: weight self-update lineage; group-wise advantage redistribution (MiMo-style); online test-time RL; self-distilled SFT. Asset-library dose–response: frozen model, fixed parametric production instances (4 × 3 rounds), library accumulated across generations, 3 seeds. Experience bank: advantage-weighted logit modulation, memory accumulating across sessions from

empty, versus a no-memory control (n = 10 per arm). Memorization decomposition: similarity-screening regularization rejects memories with self-similarity ≥ threshold against the live benchmark instance.

Threshold control. The identical inference-time stack (best-of-three R1 CAD-screened selection, in-session capital reuse, syntax repair, conditional resampling with second retry, judge-retry) was run on the untrained base weights (no RL adapter), n = 5 sessions, single-judge protocol. The experience bank starts empty and accumulates only the arm's own rating-≥6 designs (self-mined semantics — the V2 arm's bank is likewise mined from its own lineage's runs). The generation budget is raised to 2,000 tokens for all rounds (the stack default is 900 for R1 candidates and 1,500 later), because the base model's pretty-printed JSON truncates at 900; a truncation-crippled control would not test the reuse mechanisms.

Statistics. Two-sided or one-sided Mann–Whitney U as stated; Cohen's d; Spearman ρ; ordinal Krippendorff's α; bootstrap 2,000 draws. No multiple-comparison correction beyond the reported bounds; all per-session data in the repository.

Reproducibility. All code (benchmark, training, inference stack, analysis), 59 result datasets, per-run session artifacts, judge rating caches and figure-generation scripts are open-sourced. No human subjects were involved; all ratings are machine-generated.

Data availability All benchmark code, evaluation results and datasets are available at the MechanogenesisBench repository. **Code availability** Benchmark, training and inference-stack code accompany the paper in the same repository. **Acknowledgements** To be completed. **Author contributions** E.W. designed the benchmark, ran all experiments and wrote the paper. **Competing interests** The author declares no competing interests.

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Method-adaptation sources released with this work: JitRL (arXiv:2601.18510 adaptation), RRSI regularization (arXiv:2609.24972 adaptation).

Extended Data (summary)

Extended Data Fig. 1 | v1→v7 stack ablation. Mean profit by mechanism version (¥72 → ¥206 → ¥350 → ¥368 → ¥440/¥436), five to twenty sessions each.

Extended Data Fig. 2 | Mechanism comparison. CARE/GAR/TTRL/self-SFT/skill-library arms with round-level sell rates.

Extended Data Fig. 3 | CAD world model. Prediction adapter reaches 92% validity, 69% exact rating; ranker-based screening is negative (learning-to-rank fix documented).

Extended Data Table 1 | Full 39-system leaderboard with protocols, n, and per-run spreads (repository).

Extended Data Table 2 | Sealed-probe generalization. V2 stack ¥341 versus plain weights ¥327 (n = 10 per arm); round-level ratings.

Extended Data Table 3 | Threshold control: 2×2 factorial. Zero-shot base ¥0; base + identical inference-time stack ¥154 (n = 5 sessions: 240, 150, 80, 150, 150; 6/25 rounds sold; s.d. ¥57); trained weights without stack ¥390; trained weights + stack ¥438 single-judge (¥532 three-judge).

Main text ~3,300 words (excluding Methods, references and figure captions). Supplementary Information: propositions P1–P3 with proofs; formal-model simulation code.